La Scuola d'Italia / Dr. Huson / IB Math
29 May 2026

First & Last name:

8.6 Early Finishers: Transformations of Quadratics and a Cubic

Answer on separate lined or graph paper, showing full working. A graphing calculator is allowed to check your sketches, but show the algebra. The parent functions are $f(x) = x^2$ *and* $f(x) = x^3$.

1. Let $f(x) = x^2$. The function g is defined by $g(x) = -2(x+3)^2 + 5$.
   (a) Describe fully the sequence of transformations that maps the graph of f onto the graph of g.
   (b) Write down the coordinates of the vertex of g and the equation of its axis of symmetry.
   (c) State the range of g.
   (d) Express $g(x)$ in the form $ax^2 + bx + c$.
   (e) Sketch the graph of g, labelling the vertex and the y-intercept.

2. The graph of $y = x^2$ undergoes the following transformations, in order: a vertical stretch with scale factor $\frac{1}{2}$, then a reflection in the x-axis, then a translation by the vector $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.
   (a) Find an equation for the resulting function h in vertex form.
   (b) Write down the coordinates of the vertex of $y = h(x)$, and determine whether the graph has any x-intercepts, justifying your answer.
   (c) Does the order of the stretch and the reflection matter? Justify your answer.

3. Let $f(x) = x^3$. The function g is defined by $g(x) = 2(x+1)^3 - 16$.

   (a) Describe fully the sequence of transformations that maps the graph of f onto the graph of g.

   (b) Write down the coordinates of the point of inflection of the graph of g.

   (c) Find the coordinates of the x-intercept and of the y-intercept of the graph of g.

   (d) Sketch the graph of g, showing the point of inflection and both intercepts.

4. **Challenge.**

   (a) The graph of $y = x^2$ is transformed into the graph of $p(x) = -x^2 + 8x - 13$. By completing the square, describe fully the sequence of transformations, and state the coordinates of the vertex of p.

   (b) A curve has equation $y = a(x-h)^3 + k$. Its point of inflection is at $(-1, 2)$, and it passes through the point $(1, 18)$. Find the values of a, h, and k, and hence write down the equation of the curve.